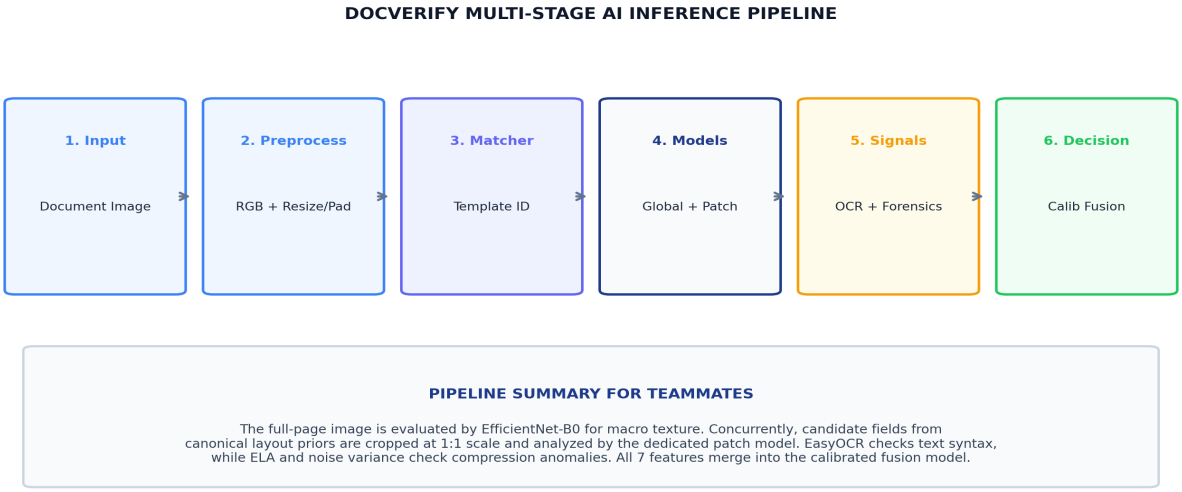
DocVerify — Phase 3: Explainability & Spatial Distortion Audit

Official Technical Milestone Report & Engineering Specification (Phase 3)

1. Phase Executive Summary & Objectives

Phase 3 investigated model interpretability using Grad-CAM attention heatmaps. The investigation revealed a critical computer vision limitation: resizing rectangular identity documents (1.6:1 aspect ratio) into standard 224x224 squares distorted the spatial gradient projections, misdirecting attention maps away from actual tampered fields.

2. Architectural Diagram & Pipeline Workflow



3. Technical Specifications & Milestone Deliverables

Dimension / Metric	Implementation / Specification	Significance & Impact
XAI Technique	Gradient-Weighted Class Activation Mapping (Grad-CAM)	Visualized intermediate activation maps at features[8] to explain model focus zones.
Spatial Distortion Issue	Squaring Rectangular Inputs (e.g. 2167x1360 to 224x224)	Caused Grad-CAM heatmaps to stretch horizontally, pointing to incorrect text areas.
Inverse Mapping Math	CorrectedGradCAM Inverse Coordinate Projection	Engineered custom matrix transformation to map heatmaps accurately back to original image space.
HiRes-CAM Addition	High-Resolution Class Activation Mapping	Element-wise gradient product implemented for finer spatial attribution.

4. Key Discoveries, Challenges & Next-Phase Directives

- Standard naive Grad-CAM is invalid for rectangular identity documents without aspect-ratio correction.
- Engineered src/gradcam_corrected.py with exact mathematical coordinate back-projection.
- Discovered that global attention often looked at borders rather than small tampered text characters.

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